

1) nano input.txt

```
hello world
hello spark
spark is fast
world of big data
big data is powerful
apache spark is popular
scala works with spark
hello big data world
spark processes data quickly
```

2) nano word_count.scala

```
val data=sc.textFile("input.txt")
data.collect;
val splitdata = data.flatMap(line => line.split(" "));
splitdata.collect;
val mapdata = splitdata.map(word => (word,1));
mapdata.collect;
val reducedata = mapdata.reduceByKey(_+_);
Reducedata.collect;
```

3) nano factorial.scala

```
def factorial(n: Int): Int = {
  if (n == 0) 1 else n * factorial(n - 1)
}

val num = 5
println("Factorial = " + factorial(num))
```

4) nano bubblesort.scala

```
def bubbleSort(arr: Array[Int]): Array[Int] = {  
  val n = arr.length  
  for (i <- 0 until n) {  
    for (j <- 0 until n - i - 1) {  
      if (arr(j) > arr(j + 1)) {  
        val temp = arr(j)  
        arr(j) = arr(j + 1)  
        arr(j + 1) = temp  
      }  
    }  
  }  
  arr  
}  
  
val arr = Array(5, 2, 9, 1, 3)  
val sorted = bubbleSort(arr)  
  
println("Sorted Array: " + sorted.mkString(", "))
```

5) spark-shell

6) :load word_count.scala

7) :load factorial.scala

8) :load bubblesort.scala